

Let's talk about the weather

X19 (2019) — English translation

The map below shows isobars (for the same altitude, approximately at sea level). Assume that the isobar diagram is stationary (changing slowly over time).

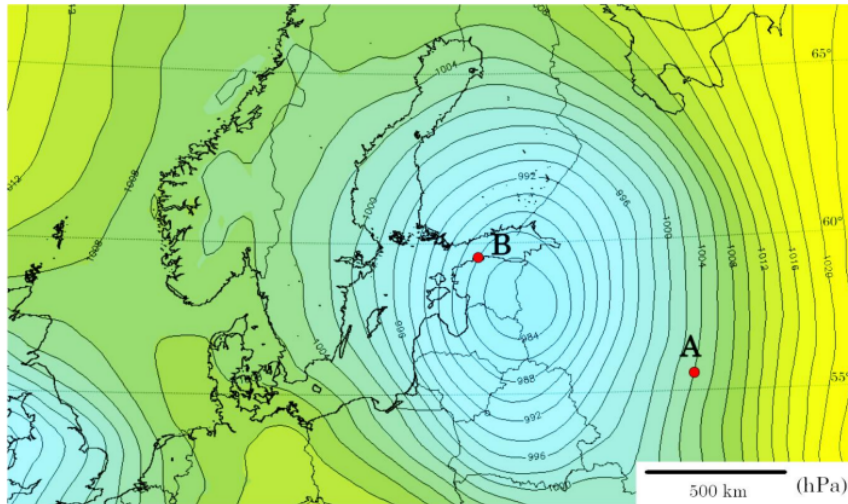


Figure 1: Figure 1.

A1	Sketch the wind direction at points <i>A</i> and <i>B</i> .	1.00
A2	Estimate the wind speed v_A at point <i>A</i> . Consider the isobars to be straight lines around this point. Air density $\rho = 1.23 \text{ kg/m}^3$.	2.00
A3	Estimate the wind speed v_B at point <i>B</i> .	3.00

Note: When a body of mass m (for example, a layer of air) moves with speed v in a rotating frame of reference (rotating with angular speed Ω), it is acted upon by a certain fictitious force, which is called the Coriolis force:

$$F_C = 2m\Omega v \sin \varphi$$

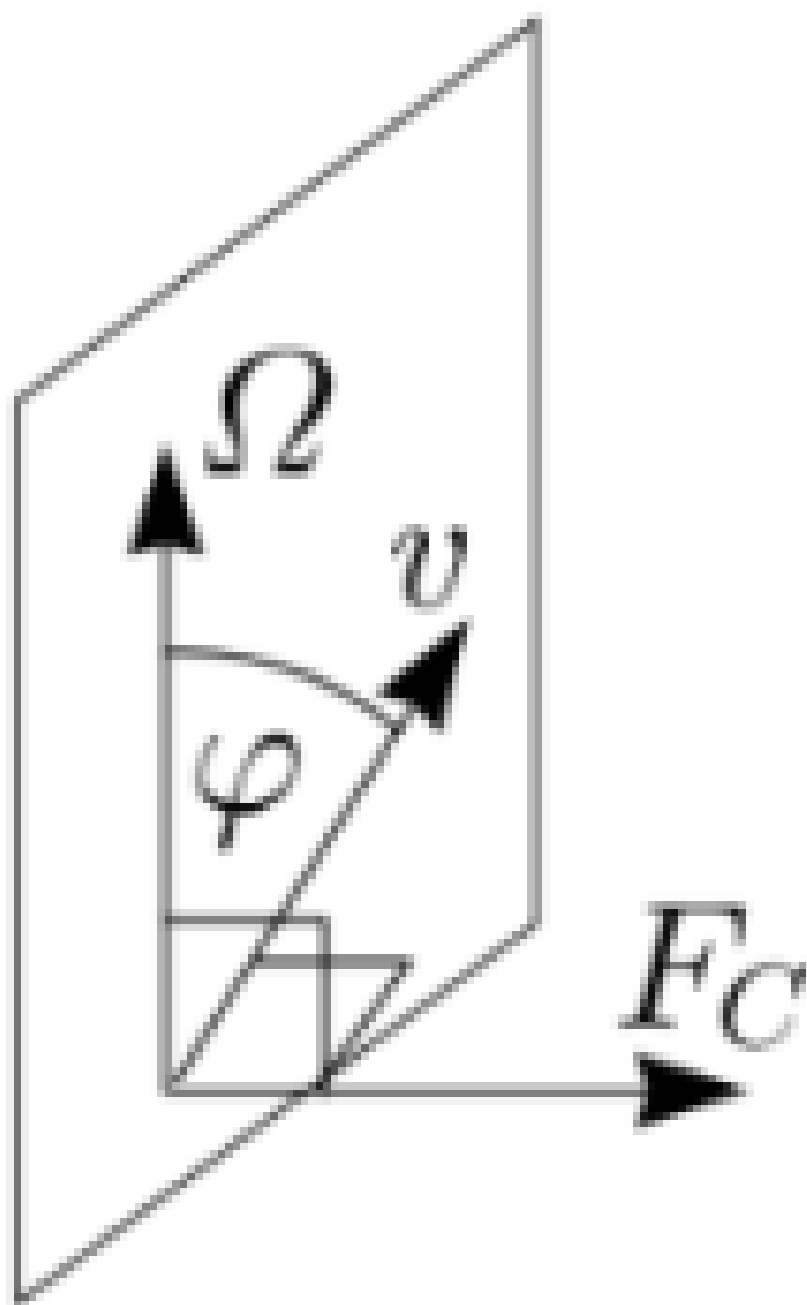


Figure 2: Figure 2.

Source: <https://pho.rs/p/3056>